

6DoF Optoelectronic Sensing for Miniaturized Soft Robotic Spines

Framework: Inverse Kinematics, Noise Modeling, and Geometric Optimization

1 The Forward Kinematics (Sensor-to-Strain)

The mapping from 6 raw sensor values (horizontal h_i and vertical v_i intensity changes) to the Cartesian 6DoF vector $\vec{O}$ is governed by the linear transformation:

$$\vec{O} = \mathbf{M}\vec{S} \quad (1)$$

Where $\vec{O} = [X, Y, Z, R_x, R_y, R_z]^T$ and $\vec{S} = [h_1, v_1, h_2, v_2, h_3, v_3]^T$. For a triangular array with radius L , the matrix $\mathbf{M}$ is defined by the geometry:

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & -0.5 & 0 & -0.5 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{2} & 0 & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & \frac{1}{L} & 0 & -\frac{0.5}{L} & 0 & -\frac{0.5}{L} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{2L} & 0 & -\frac{\sqrt{3}}{2L} \\ \frac{1}{L} & 0 & \frac{1}{L} & 0 & \frac{1}{L} & 0 \end{bmatrix} \quad (2)$$

Note on Scaling: As $L \rightarrow 0$ (miniaturization), the rotational terms ($\frac{1}{L}$) approach infinity, meaning a tiny bit of noise in the light intensity is amplified into a massive rotational error.

2 The Inverse Problem (Synthetic Data Generation)

To simulate what the sensors "see" for a given robot pose, we invert the system:

$$\vec{S} = \mathbf{M}^{-1}\vec{O} \quad (3)$$

```
1 % Generate sensor intensities for a 5-degree Pitch (Rx)
2 strain_vector = [0; 0; 0; 5; 0; 0];
3 synthetic_sensors = M \ strain_vector; % Backslash is the inverse
   solver
```

Listing 1: Inverse Solver Snippet

3 Stochastic Noise Modeling

In your 3mm board, the signal-to-noise ratio (SNR) is dominated by Shot Noise and Quantization. We model the reconstructed error Σ_O based on input sensor variance σ_s^2 :

$$\Sigma_O = \mathbf{M} \begin{bmatrix} \sigma_{s1}^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{s6}^2 \end{bmatrix} \mathbf{M}^T \quad (4)$$

The Condition Number (κ): A key metric for optimization is $\kappa(\mathbf{M})$. If $\kappa(\mathbf{M})$ is high, the matrix is "ill-conditioned," and the 6DoF output becomes unstable.

4 Spacing and Saint-Venant's Localization

The legitimacy of a localized strain measurement is constrained by the mechanical interaction between the rigid sensor and the soft substrate.

- **Influence Zone (R_{inf}):** Dissipates at a distance $d \approx 2 \times D_{sensor}$.
- **Reconstruction Density:** Spacing S must satisfy the Nyquist-like criterion:

$$S < \frac{\rho_{min}}{2} \quad (5)$$

```

1 % Loop to find the "Knee" of the error curve
2 for L = linspace(0.5, 10, 100)
3     Matrix_Condition(end+1) = cond(build_M(L));
4 end
5 plot(L, Matrix_Condition); % Identify physical limit

```

Listing 2: Condition Number Optimization